



First Semester 2026-2027
Course Handout (Part II)

Date: 02/08/2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific detail regarding the course:

Course No: **EEE F426**
Course Title: **Fiber Optics and Optoelectronics**
Instructor-in-charge: **VINOD KUMAR CHAUBEY (vkc@pilani.bits-pilani.ac.in)**

1. Course Description:

Optical communication systems and components; optical sources and transmitters (basic concept, design and applications); modulators (electro- optic, acousto-optic and laser modulation techniques); beam forming; focusing and coupling schemes to optical repeaters; optical amplifiers; optical field reception; coherent and non-coherent light wave systems; fiber optic communication system design and performance; multichannel light wave systems; long haul communications; fiber optic networks.

2. Scope & Objective:

In the recent past, tremendous advances have been achieved in fiber optics and associated optoelectronics. These developments have made fiber - optic communication synonymous with the current worldwide revolution in information technology. This course aims at providing the students with a firm grounding in the major aspects of this emerging technology. Thus the course deals with the study of various building blocks of fiber optic systems, e.g. optical fibers, sources, detectors, modulators, optical amplifiers, etc. together with overall system design and performance analysis for communication applications.

3. Course Learning outcomes:

At the end of this course, students will be able to:

- **CLO1:** Basic configuration of a fiber-optic system, Merits of Fiber Optic system, Ray propagation through different types of optical fibers and Estimation of different parameters of optical fibers.
- **CLO2:** Familiarization with the background for learning of electromagnetic wave propagation in medium, Modal analysis of planar optical wave guides, Modal analysis of cylindrical optical waveguide and characterization of optical fibers.
- **CLO3:** Concept of semi-conductor p-n junction towards design LED, LED designs characterization and ILD condition for laser action, Optical transmitter design and applications
- **CLO4:** Basic principle of opto-electronic detection, Types of photodiodes, Structures of photodetectors and optical receivers, Operation of Optical Amplifiers, Optical modulators and demodulators.
- **CLO5:** Optical Communication system design for single wavelength and multi wavelength systems. Design of fiber optic sensors, Extrinsic and Intrinsic Sensors Distinct Types of Sensors and their Industrial Applications

4. Text Book (TB): Khare, R.P.: "Fiber Optics and Optoelectronics" Oxford University Press (2004)

Reference Book (RB): Keiser. G.. "Optical Fiber Communications", 5/e, McGraw Hill, 2013





5. Course Plan:

Module	Lect. No.	Topic (s) to be covered	Reference to chapter of TB/RB	Learning Outcomes
I	1,2	A generalized configuration of a fiber optic system, Advantages, Role of fiber optic systems, optical spectrum.	Chapter 1(TB, RB)	Basic configuration of a fiber-optic system, Merits of FO system, Role-play in the sociological evaluation
	3, 4	Review of fundamental laws of optics, Ray propagation in step index fibers, Ray propagation in graded index fibers, Effect of material dispersion, Effect of multipath-dispersion and combined effect, Numerical problems	Chapter 2 (TB, RB)	Ray propagation through different types of optical fibers, Estimation of causes of pulse broadening, Calculation of different parameters of optical fibers
II	5, 6, 7, 8	Maxwell's equations, Solution in an inhomogeneous medium, Planar optical waveguide, TE modes of a symmetric step index planar waveguide, Power distribution and confinement factor, Wave propagation in an ideal SI fiber, Modal power distribution in SI fibers, Wave propagation in GI fiber	Chapter 2 (RB), Chapter 3 (TB), Chapter 4 (TB)	Familiarization with the background for learning electromagnetic wave propagation, Modal analysis of planar optical wave guides, Modal analysis of cylindrical optical waveguide
III	9, 10, 11, 12	Propagation through Single mode fibers, Single mode fibers, Fabrication of low loss optical fibers, Design aspects of optical fiber cables and connections, Evaluating the performance of optical fibers	Chapter 2, 3, 14 (RB)	Characteristic parameters of SM fibers, Dispersion in SM fibers, Attenuation in SM fibers, Design of SM fibers and related problems, Fiber material requirements, Fiber fabrication methods, Fiber optic cables, connection and related losses, Characterization of optical fibers
IV	13, 14, 15, 16	Fundamental aspects of optoelectronic sources, Principle of operation of light emitting diodes(LED), Design aspects of LED, Principles of injection laser diode (ILD), Design aspects of ILD and source fiber coupling.	Chapter 4, 5 (RB)	Intrinsic and extrinsic semi-conductors, p-n junction, Life time and diffusion length of minority carriers, Current density and injection efficiency, LED, its internal and external quantum efficiency, Heterojunction, LED designs, ILD, condition for laser action, laser modes & laser action in semiconductors, ILD structures, Source Fiber coupling.



V	17, 18, 19, 20	Principle of operation and types of optoelectronic detectors – photodiodes, phototransistors, APDs, Detector Noise and Response Time, Optical Receiver Operation	Chapter 6, 7 (RB)	Basic principle of opto-electronic detection, Types of photodiodes, Structures of photodetectors, Analog and Digital Receiver Operation.
VI	21, 22, 23, 24	Review of basic principles of optoelectronic modulator, Electro optic effect and related modulators, Acousto- optic effect and related modulators	Chapter 9 (TB)	Polarization, birefringence, retardation plates, Electro optic modulators and related problems, Acousto-optic modulators & related problems.
VII	25, 26, 27, 28	Optical amplification and amplifiers – SOA, EDFA, SRS, Raman amplifier, amplifier noise, SNR	Chapter 11 (RB)	Semiconductor optical amplifiers, Erbium doped fiber amplifiers, Fiber Raman amplifiers
VIII	29, 30, 31, 32	Analog and Digital Links, WDM Components, Optical communication systems and networks, Multichannel Transmission, DWDM, SONET/SDH, High speed links, Free space optic channel or optical wireless communication	Chapter 8, 9, 10, 13 (RB)	Concepts of WDM & DWDM, Passive and active components, System design considerations, System architectures, Network concepts and topologies, PON and Ethernet, RF over Fiber, IP over DWDM, FSO/OWC.





IX	32,34, 35,36	Fiber Optic Non-Linearities, SBS, SRS, SPM, XPM, FWM	Chapter 12 (TB & RB)	Overview on non-linearities, effective fiber length and cross-sectional area, nonlinear effects and their mitigation, dispersion management.
X	37,38, 39,40	Introduction to Fiber optic Sensors, Intensity-, Phase-, Spectrally-Modulated Sensors, Distributed Fiber-optic Sensors.	Chapter 13 (TB)	Sensors based on fiber optic cable, sensing principles, Extrinsic and Intrinsic Sensors Distinct Types of Sensors, Industrial Applications

6. Evaluation Scheme:

EC. No.	Evaluation Component	Weightage (%)	Duration	Date, Time & Venue	Nature of Component
1.	Mid Semester Test	30	90 min.	06/10/2026 AN1	Closed Book
2.	Quiz (Announced or Surprise) / Class participation credit	20	(10 min) beginning / before the end of the class		Closed
3.	Assignment/ Interaction	10	TBA on Nalanda Course page		Open Book
4.	Comprehensive Exam	40	3 hrs.	07/12/2026 FN	Closed/Open Book

7. Consultation Hour: Thursday at 4.00 PM, **Room no: 2243-E.**

8. Notices: Notices concerning this course will be displayed on **Nalanda Course-page.**

9. Make-up Policy: Make-up in Tests/ Comprehensive Examination shall be given for genuine reasons (e.g. medical, personal emergencies, etc.) on submission of documents, only if applied in advance to scheduled test/exam. **No make-up is allowed for quiz component** due to buffer quizzes as applicable.

Instructor-in-Charge

EEE F426

